# Multi-spin quenches: where is an RBM predicted easy and an MPS hard?

Exact-diagonalisation screen (RBM runs to follow)

30 October 2026

**Abstract**

Are there quenches with three- and four-spin interactions in which a restricted Boltzmann machine (RBM) is predicted to follow the exact state while a matrix product state (MPS) would need a large bond dimension? We evolved seven families on 12–18 spins exactly and computed four RBM predictors (few-body error rate $R_2$, its integral, high-order cluster weight $w_{hi}$, cumulant level $\mu$) and the entanglement entropy $S$. The predictors disagree. $\mu$ predicts an easy RBM at MPS-hard times for the toric code in a diagonal field (16–24% of the save times for $h_f \leq 0.4$, 18 spins) and for the Baxter–Wu and Newman–Moore quenches at weak field (19% and 31% at $h = 0.05$, below 5% for $h \geq 0.2$); $w_{hi}$ calls all of these hard within 0.3 time units. All other product-state quenches are hard by every predictor. On two reference quenches with RBM data, $\mu$ tracks the infidelity best (Spearman 0.91, 0.93) but can be late by up to 2.6×. RBM runs are needed to decide.

## 1 Overview

**Question.** The previous report on two-spin quenches found an RBM advantage over MPS only in near-classical quenches, and argued that three- and four-spin terms freeze an RBM that starts from a product state. Are there multi-spin quench families, with a control parameter, in which some quenches are predicted easy for an RBM while the state is MPS-hard, and others hard?

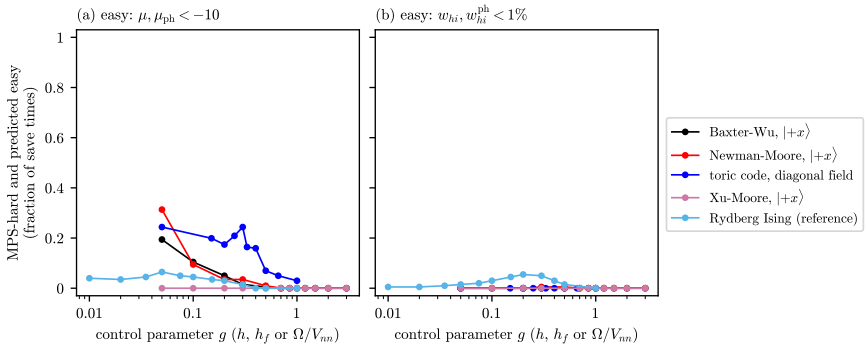



**Figure 1: Graphical abstract.** Fraction of the save times at which the exact state is MPS-hard (entanglement entropy $S \geq S_{\rm Page}/2$, half the value of a random state) *and* still predicted easy for an RBM, against the control parameter $g$ of each family (transverse field $h$, final field $h_f$, or drive $\Omega/V_{nn}$; log scale). (a) "Easy" means before both cumulant levels first exceed the threshold, $\mu, \mu_{\rm ph} < -10$. (b) "Easy" means before both high-order weights first exceed 1%, $w_{hi}, w_{hi}^{\rm ph} < 0.01$. Families: Baxter–Wu ($-\sum_{\triangle} ZZZ - h\sum X$, 18 spins) and Newman–Moore ($-\sum_{\triangle_\uparrow} ZZZ - h\sum X$, 16 spins) from $|+x\rangle$; toric code $-\sum_s A_s - \sum_p B_p - h_f \sum(Z+X)/\sqrt{2}$ from its ground state at $h_f = 0.1$ (18 spins); Xu–Moore ($-\sum_{\square} ZZZZ - h\sum X$, 16 spins) from $|+x\rangle$; the two-spin Rydberg Ising quench of the previous report as reference (12 spins). Not shown: the no-quench point $h_f = 0.1$ and the points at $g = 0$ (log axis; see text), the mixed families and the ground-state starts of the plaquette and triangle models (Section 6). Run lengths: $t_{\max} = \min(300, \max(10, 3/h))$ for the $|+x\rangle$ starts, 10 for the toric code.

**Method** (Sections 2 and 3). Exact evolution of 7 families (B1–B7, Table 1) and 4 reference families on 12–18 spins, 12 values of the control parameter per family (8–13 for the references), 186 runs in total. From the exact state $\psi(s,t)$ we compute the predictors of Section 2 at 201 times per run. No RBM was run in this study.

**Answer** (Sections 4 to 7).

- *Predicted easy at MPS-hard times, by $\mu$ only:* the **toric-code quench** for $h_f \leq 0.4$ (Section 4), and the **Baxter–Wu quench** and **Newman–Moore quench** at weak field, with windows that shrink steadily as $h$ grows (Section 5).
- *The predictors disagree on exactly these cases.* The high-order weight $w_{hi}$ crosses 1% within $t \approx 0.3$, and $R_2$ stays between 4.9 and 6 in the Baxter–Wu quench.
- *Hard by every predictor:* from $|+x\rangle$, Xu–Moore and the two mixed families, within $t \leq 0.6$ (Section 6). Of the ground-state starts of the plaquette and triangle models, only Newman–Moore opens small $\mu$-windows.
- *Two rows give no usable prediction:* the toric code in a $Z$ field (B1) stays in a sector of 256 of $2^{18}$ configurations, so all predictors computed in the $Z$ basis are meaningless; and the ground-state start of Baxter–Wu turned out to be an ordered cat state (Appendix F).

## 2 Predictors

All quantities refer to $N$ spins in the $Z$ basis, $s \in \{\pm 1\}^N$, and the cluster functions $s_A = \prod_{i\in A} s_i$ for a set $A$ of sites ($|A|$ is the order of the cluster). Energies are in units of the multi-spin coupling ($J = 1$), times in units of $1/J$.

**Definition: few-body error rate $R_k$ and its integral $\Phi_k$**

Write the exact evolution as $\partial_t \log\psi(s) = v(s)$ with the local velocity $v(s) = -i\,(H\psi)(s)/\psi(s)$. The few-body error rate is the part of $v$ that no $k$-body function can match, measured with the Born weight $p(s) = |\psi(s)|^2$:

$$R_k^2 = \min_c \sum_s p(s) \left| v(s) - \sum_{|A|\leq k} c_A s_A \right|^2,$$

with complex $c_A$ for all clusters up to $k$ sites (all-to-all). $R_2 = 0$ means that a two-body Jastrow factor can follow the evolution exactly for an instant. The accumulated error is $\Phi_k(t) = \int_0^t R_k\, dt'$. Implementation and checks: Appendix D.

$R_2$ is the main variant. It was validated only on the transverse-field Ising chain, where an RBM represents two-body phases exactly. For multi-spin $H$ it measures the distance to a two-body motion, which is stricter than what an RBM with weights of order one can do; $R_4$ (clusters up to 4 sites) is shown in the appendix. $R_2$ and $\Phi_2$ have no threshold here, because the size rescaling of the published threshold could not be checked; they are read only as "small" or "large".

**Definition: cluster predictors $\mu$ and $w_{hi}$**

- $c_A = 2^{-N}\sum_s s_A \log|\psi(s)|$: cluster coefficients of the log-amplitude. Coefficients with $\ln|c_A| < -28$ count as numerical noise and are dropped.
- *Cumulant level $\mu$*: rank the remaining $N_{\text{real}}$ coefficients by size and fit the rank as a function of $x = \ln|c_A|$ with the Fermi–Dirac form $N_{\text{real}}/(e^{(x-\mu)/T}+1)$. So $\mu$ is a natural logarithm, and about half of the coefficients are larger than $e^{\mu}$: $\mu = -10$ means that

the typical relevant cluster has $|c_A| \approx e^{-10} \approx 5 \times 10^{-5}$. Small $\mu$ means few relevant clusters. If no cluster beyond two sites survives the noise cut, $\mu = -\infty$ (shown at $-30$). $\mu_{\mathrm{ph}}$ is the same for the phase $\arg\psi$, unwrapped in time, with its mean removed.

- *High-order weight* $w_{hi} = \sum_{|A|>4} c_A^2 / \sum_{|A|\geq 1} c_A^2$, and $w_{hi}^{\mathrm{ph}}$ for the phase. We set $w_{hi} = 0$ when the denominator is below $10^{-20}$ (flat amplitude, e.g. a product state; without this the ratio of two round-off numbers exceeds 1%).

$\mu$ measures the *absolute* size of the many-body clusters, $w_{hi}$ their weight *relative* to all clusters.

**Entanglement.** $S$: von Neumann entropy (nats) of the exact state between the first $N/2$ sites and the rest. On the square and triangular tori the halves are two blocks of columns; on the toric-code tori they are the links with midpoint $x \leq 1$ against the rest, so the cut is spatial (Appendix B). *MPS-hard* means $S \geq S_{\mathrm{Page}}/2 = \frac{1}{2}(\frac{N}{2}\ln 2 - \frac{1}{2})$: 1.83, 2.52 and 2.87 nats for $N = 12$, 16, 18.

**Definition: onset times and the predicted window**

- $t_\mu^*$: first time with $\mu > -10$ or $\mu_{\mathrm{ph}} > -10$.
- $t_w^*$: first time with $w_{hi} > 1\%$ or $w_{hi}^{\mathrm{ph}} > 1\%$.
- $t_S$: first time at which the state is MPS-hard.
- *Predicted window* of a criterion: the fraction of the 201 save times at which the state is MPS-hard *and* $t < t^*$ ($t^* = t_\mu^*$ or $t_w^*$); 0 if the state is never MPS-hard. At weak fields $S$ oscillates around the threshold, so this is smaller than the time between $t_S$ and $t^*$.

The non-classicality $\eta = \|H - \mathrm{diag}H\|_F / \|H - \mathrm{tr}H/2^N\|_F$ is 0 for a diagonal (Ising) Hamiltonian and 1 for a purely off-diagonal one.

## 3 Setup of the highlighted quenches

The main text discusses three cases; Table 1 lists all families.

- **toric-code quench** (B2): $H = -\sum_s A_s - \sum_p B_p - h_f \sum_i (Z_i + X_i)/\sqrt{2}$, with star $A_s = \prod_{i\in s} X_i$ and plaquette $B_p = \prod_{i\in p} Z_i$, on the links of a $3 \times 3$ torus (18 spins; also $2 \times 3$, 12 spins), from the ground state at $h_f = 0.1$ (gap $1.2 \times 10^{-3}$). The diagonal field makes neither $A_s$ nor $B_p$ conserved. We know of no duality that maps this case to a two-spin model; the $Z$-field case maps to the transverse-field Ising model [1, 2]. $\eta = 0.71$.
- **Baxter–Wu quench** (B4): $H = -\sum_\triangle Z_i Z_j Z_k - h \sum_i X_i$ on all triangles of an 18-site triangular torus ($6 \times 3$, compatible with the three-sublattice order of the model [3]), from $|{+x}\rangle$, $\eta = 0.04$–$0.91$. The **Newman–Moore quench** (B5) [4], with up-triangles only, on a $4 \times 4$ torus, behaves similarly and is quoted alongside.
- **Xu–Moore quench** (B3): $H = -\sum_\square Z_i Z_j Z_k Z_l - h \sum_i X_i$ on a $4 \times 4$ torus [5], from $|{+x}\rangle$. It stands for the families that are hard everywhere.

The final field $h_f$ is the control parameter of the toric code (0–1; $t_{\max} = 10$), and the field $h$ that of the $|{+x}\rangle$ starts (0.05–3; run length $t_{\max} = \min(300, \max(10, 3/h))$).[1]

[1] Exact evolution with Krylov exponentials (`expm_multiply`) in steps of at most $\min(\Delta t_{\mathrm{save}}, 0.1/\|H_{\mathrm{off}}\|_\infty)$, 201 save times per run. The phase is unwrapped exactly for the diagonal part and by step-to-step continuity for the rest. Cost and data: Appendix G.

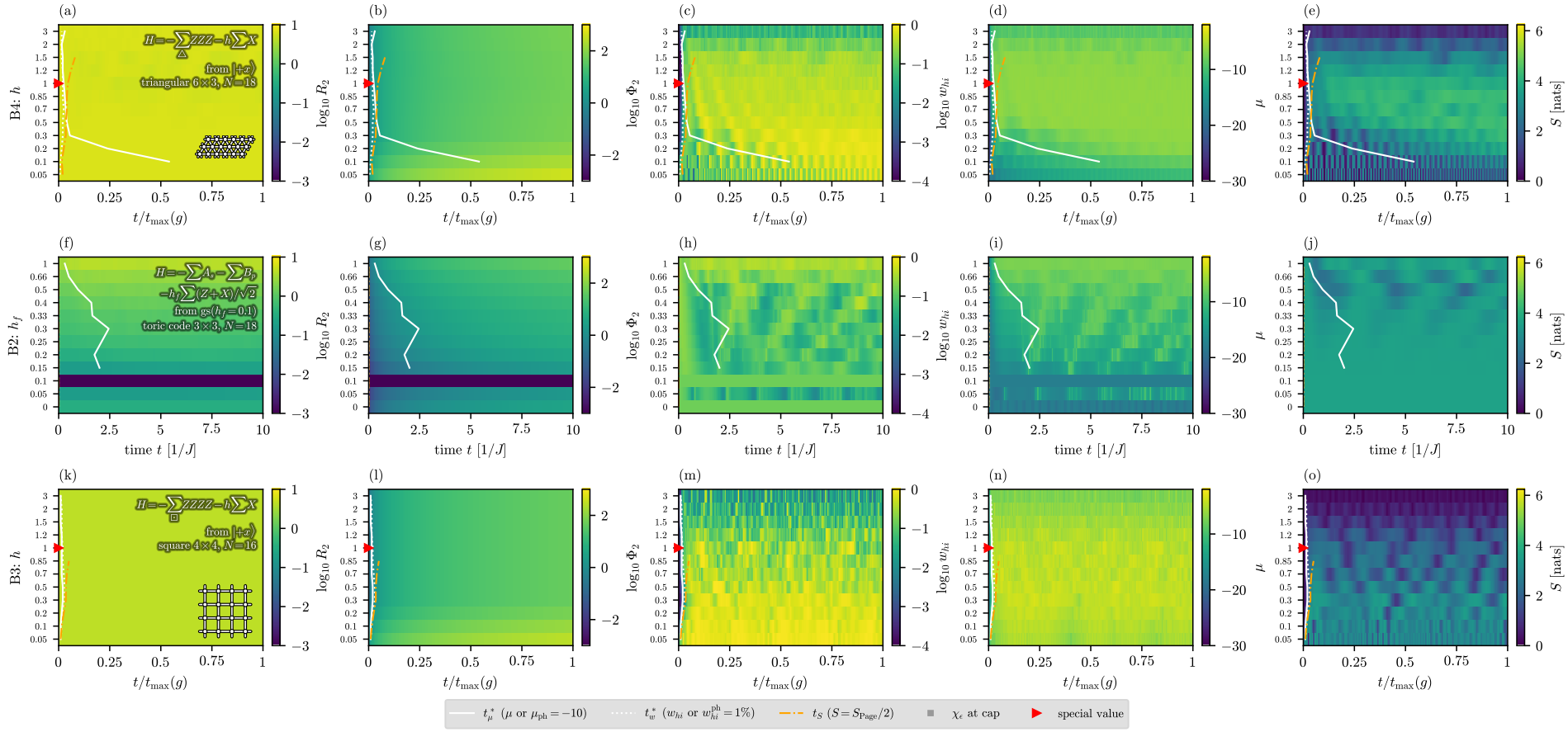



**Figure 2: Predictors of the three highlighted quenches over time $t$ (horizontal) and control parameter (vertical, one row per grid point).** Rows of panels: (a)–(e) **Baxter–Wu quench** from $|+x\rangle$, 18 spins, time in units of the run length $t_{\max}(h) = \min(300, \max(10, 3/h))$; (f)–(j) **toric-code quench** from the ground state at $h_f = 0.1$, 18 spins; (k)–(o) **Xu–Moore quench** from $|+x\rangle$, 16 spins, time in units of $t_{\max}(h)$. Columns: few-body error rate $\log_{10} R_2$, its integral $\log_{10} \Phi_2$, high-order weight $\log_{10} w_{hi}$, cumulant level $\mu$ (floor $-30$ = exactly two-body), entanglement entropy $S$ in nats (grey squares: times at which an MPS with discarded weight $10^{-3}$ needs the largest possible bond dimension at the cut with the largest $S$). Lines on every panel: solid white, $t^*_\mu$ ($\mu$ or $\mu_{\text{ph}}$ first above $-10$); dotted white, $t^*_w$ ($w_{hi}$ or $w^{\text{ph}}_{hi}$ first above 1%); dash-dotted orange, $t_S$ (first $S \geq S_{\text{Page}}/2$: 2.52 nats for 16 spins, 2.87 for 18). A line at the left edge means an onset at $t = 0$ (all $t^*_w$ and $t_S$ of the toric code); a missing line means no onset within the run. Red triangles: self-dual point $h = 1$ (assumed for Baxter–Wu, see Appendix F). Insets: Hamiltonian, start state, lattice (sketch omitted for the toric code). The toric-code row at $h_f = 0.1$ is the start state itself (no quench).

## 4 Toric code in a diagonal field: the only non-classical $\mu$-window

*The **toric-code quench** is the only family with $\eta \approx 0.7$ in which $\mu$ predicts an easy RBM at MPS-hard times, and its windows are the largest of the non-trivial quenches* (Fig. 2f–j, Fig. 1).

**The window.** For $h_f = 0.05$–$0.4$ (all quenches with $0 < h_f \leq 0.4$; $h_f = 0.1$ is no quench), $t^*_\mu = 1.6$–$2.45$ at 18 spins and 0.45–2.35 at 12 spins. The window is 16–24% of the save times at 18 spins and 5–23% at 12 spins. For $h_f \geq 0.5$ it shrinks to 3–7%. $\mu_{\text{ph}}$ crosses first in most rows; $\mu$ stays near $-15$ for $h_f \leq 0.15$ up to $t = 10$. $R_2$ grows smoothly with $h_f$, from 0.2 ($h_f = 0.15$) to 3.7 ($h_f = 1$) at late times, without a sharp change.

**The quench to the pure toric code.** At $h_f = 0$, $\mu$ and $\mu_{\text{ph}}$ stay below $-10$ for the whole run (window 100%). The final Hamiltonian is a sum of commuting stabilisers, much like the classical quench limits of the two-spin families, so we do not count this point as evidence.

**The entanglement is there from the start.** The state is MPS-hard at $t = 0$: $S = 3.59$ nats at 18 spins, above the threshold 2.87, and it barely changes during the run. We attribute this to the area-law entropy of the topological ground state across a cut that crosses the torus twice; it is not generated by the quench. The window thus measures how long the quench keeps the state simple, not how fast it becomes complex.

**The high-order weight disagrees.** $w_{hi} = 0.13$ already at $t = 0$, so $t^*_w = 0$ for every $h_f$. The start state is close to the toric-code ground state, which has an exact RBM with $\alpha = 1$ [6], and $\mu = -18$ at $t = 0$. We read this as a failure of $w_{hi}$ for this class of states, but did not test it.

**Finding: toric code in a diagonal field**

From the toric-code ground state, $\mu$ predicts an easy RBM for 16–24% of the save times ($t \lesssim 2.5$) for $0 < h_f \leq 0.4$ at 18 spins (5–23% at 12 spins), at $\eta = 0.71$. The state is MPS-hard throughout because the start state is already entangled. $w_{hi}$ calls even the start state hard, although that state is close to one with an exact RBM.

## 5 Weak-field three-spin quenches: $\mu$ says easy, $w_{hi}$ says hard, $R_2$ is large

*At weak fields the **Baxter–Wu quench** and the **Newman–Moore quench** are predicted easy by $\mu$ long after the state first becomes MPS-hard; $w_{hi}$ calls them hard almost at once, and $R_2$ stays large* (Fig. 2a–e, Fig. 1).

**The cumulant level.** In the Baxter–Wu quench, $\mu$ and $\mu_{\text{ph}}$ stay below $-10$ until $t^*_\mu = 3.6$ ($h = 0.2$), 16.2 ($h = 0.1$) and beyond $t_{\max} = 60$ ($h = 0.05$). The state first becomes MPS-hard at $t_S = 0.4$–$1.2$, but at $h \leq 0.1$ its entropy oscillates and exceeds the threshold 2.87 nats only intermittently (largest $S$: 3.1–3.2 nats; Fig. 2e). The window is therefore 5%, 10% and 19% of the save times (at $h = 0.2$, 0.1, 0.05), and below 2% for $h \geq 0.3$. The Newman–Moore quench gives 31%, 10%, 3.5%, 3.5% and 1% at $h = 0.05$, 0.1, 0.2, 0.3, 0.5. The windows shrink steadily with $h$; there is no sharp boundary.

**The few-body error rate.** $R_2$ starts at 6 for every $h$ and stays between 4.9 and 6 for the whole run (Fig. 2a). The value is analytic: from $|{+x}\rangle$ the velocity $v(s, 0)$ contains the classical energy $\sum_\triangle s_i s_j s_k$, whose 36 triangle terms are orthogonal to all one- and two-body functions under the uniform weight, so $R_2(0) = \sqrt{36}$. Since $R_2$ is large for every $h$, it does not separate the weak-field

quenches from the others. $R_4$, which admits three- and four-spin clusters, is 0 at $t = 0$ and stays below 0.35 for $h \leq 0.1$ (Fig. 7).

**The high-order weight.** $w_{hi}$ crosses 1% at $t_w^* = 0.15$–$0.3$ for every $h$ (at $h \leq 0.1$ this is the first or second save time, so an upper bound), and reaches 43% at $t = 30$ for $h = 0.05$, while $\mu \approx -12$. We attribute the difference to what the two measure: at weak field the log-amplitude presumably varies only by $O(h)$, so a small absolute high-order part is already a large relative weight.

**Why this matters.** The phase of this state is close to $e^{it\sum_{\triangle} s_i s_j s_k}$, an exactly three-body function. An RBM with weights of order one can write three-body phases, but a TDVP run started from a product state with zero hidden biases does not move in the direction of odd three-body terms (previous report). So $\mu$ asks whether the state is *representable*, $R_2$ whether a two-body *motion* can follow it. We interpret the disagreement as this difference, but did not test it.

**Finding: three-spin quenches**

In the Baxter–Wu and Newman–Moore quenches at $h \leq 0.1$, $\mu$ predicts an easy RBM for 10–31% of the save times while the state is MPS-hard. $w_{hi}$ exceeds 1% by $t = 0.3$, and $R_2$ stays between 4.9 and 6 (Baxter–Wu). Only RBM runs can tell which predictor is right.

## 6 Hard everywhere: plaquette and mixed models

*From $|+x\rangle$, the **Xu–Moore quench** and the two mixed families are hard by every predictor within $t \leq 0.6$, at every value of the control parameter* (Fig. 2k–o, Appendix C).

- **Xu–Moore quench**: $t_\mu^* \leq 0.6$ and $t_w^* \leq 0.6$; no window at any $h$. $R_2$ is constant in time and equals the energy spread, $R_2 = \Delta E = 4$ from $|+x\rangle$: the subsystem symmetries (spin flips of a whole row or column) forbid every two-body cluster, so only the constant part of $v$ can be fitted.
- Ising plus plaquette (B6) and Baxter–Wu plus Ising (B7): $t_\mu^*, t_w^* \leq 0.5$, and $S$ reaches 4.5–5.0 nats. The added two-spin term, meant to build up RBM weights, does not open a window for any $K$ or $\lambda$.
- Ground-state starts at $h = 1.5$ (Figs. 4 and 5): Xu–Moore and Baxter–Wu have $\mu > -10$ already at $t = 0$. Newman–Moore opens $\mu$-windows of 9% and 1.5% at $h = 0.1$, 0.3 only. The quenches to the classical point $h = 0$ give $w_{hi}$-windows of about 60% in Xu–Moore and Newman–Moore, which we do not count (diagonal final Hamiltonian).

## 7 Verdict

**Answers.**

- *Are there multi-spin families with a predicted easy/hard split along the control parameter?* According to $\mu$, there are families whose window shrinks with the control parameter: the toric code in a diagonal field (16–24% for $0 < h_f \leq 0.4$, 3–7% above) and the Baxter–Wu and Newman–Moore quenches (from 19–31% at $h = 0.05$ to below 5% for $h \geq 0.2$). None has a sharp boundary. According to $w_{hi}$, the $|+x\rangle$ and toric-code quenches have no window (largest 0.5%). $R_2$ has no threshold yet, but stays large in the three-spin quenches.
- *Is any predicted window also non-classical?* Only the toric code ($\eta = 0.71$), and there the MPS-hardness comes from the start state.

- *How far can $\mu$ be trusted?* On the two reference quenches with RBM data, $\mu$ tracks the $\alpha = 1$ infidelity as well as or better than $R_2$ and $w_{hi}$ (Spearman 0.93 against 0.51 and 0.47 in the Rydberg Ising quench, all 0.91 in the transverse-field Ising quench). Its onset was 1.1–2.6 times late in the Rydberg quench but early at large fields in the transverse-field Ising quench (Appendix E). If the multi-spin quenches behave like the Rydberg quench, the $\mu$-windows above are upper bounds; this is untested.

**Recommended RBM runs.** The predictors disagree most clearly on three sets of points: the toric code at $h_f = 0.15$, 0.3 (12 spins first, then 18) and the Newman–Moore quench at $h = 0.05$, 0.1, 0.2 (16 spins), both cheap with TDVP on exact sums over all configurations; and the Baxter–Wu quench at $h = 0.05$, 0.1 (18 spins) only if that size is affordable. The three-spin quenches need non-zero initial hidden biases, otherwise the RBM does not move (previous report).

**Limitations.**

- No RBM was run; every statement about RBMs is a prediction.
- The thresholds $\mu = -10$ and $w_{hi} = 1\%$ were calibrated on two-spin quenches. Their meaning for three- and four-spin states is untested, and the two disagree here.
- 12–18 spins. The MPS-hard threshold is half the random-state entropy at that size, $e^S \approx 6$–18; this is small in absolute terms.
- The toric code in a $Z$ field (B1) and the ground-state start of Baxter–Wu give no usable prediction (Appendix F).

# A Notation

| symbol | name | defined in |
|---|---|---|
| $s$, $s_A$ | spin configuration, cluster function | Section 2 |
| $v(s)$ | local velocity $-i(H\psi)/\psi$ | Section 2 |
| $R_k$, $\Phi_k$ | few-body error rate, accumulated error | Section 2 |
| $c_A$, $\mu$, $\mu_{\mathrm{ph}}$ | cluster coefficients, cumulant levels | Section 2 |
| $w_{hi}$, $w_{hi}^{\mathrm{ph}}$ | high-order cluster weights | Section 2 |
| $S$, $S_{\mathrm{Page}}$ | half-cut entropy, random-state value | Section 2 |
| $t_\mu^*$, $t_w^*$, $t_S$ | onset times | Section 2 |
| $\eta$ | non-classicality | Section 2 |
| $\chi_\epsilon$ | MPS bond dimension for discarded weight $10^{-3}$ | Fig. 7 |
| $g$; $h$, $h_f$, $K$, $\lambda$ | control parameter; its value per family | Table 1 |
| $A_s$, $B_p$ | toric-code star and plaquette | Section 3 |
| $t_{\max}$ | run length | Section 3 |
| $I$ | RBM infidelity $1 - \lvert\langle\psi_{\mathrm{ED}}\rvert\psi_{\mathrm{RBM}}\rangle\rvert^2$ | Appendix E |

# B All families

**Lattices.** Periodic in both directions. Square and triangular tori are labelled by columns $x$ and rows $y$, site $i = xL_y + y$; the entanglement cut separates the first $L_x/2$ columns from the rest. Triangular lattice vectors $(1, 0)$ and $(\frac{1}{2}, \frac{\sqrt{3}}{2})$; up-triangles $(x, y), (x+1, y), (x, y+1)$, down-triangles $(x+1, y), (x, y+1), (x+1, y+1)$. Toric-code tori have one spin per link, ordered column by column (vertical links at $x$, then horizontal links at $x + \frac{1}{2}$); the cut runs between link midpoints $x \leq 1$ and $x > 1$ on the $3 \times 3$ torus and $x \leq \frac{1}{2}$ on the $2 \times 3$ torus.

| ID | Hamiltonian | lattice ($N$) | start | control $g$ | $\eta$ |
|---|---|---|---|---|---|
| B1 | toric code $-h_f \sum Z$ | toric $2\times3$ (12), $3\times3$ (18) | gs($h_f = 0.1$) | $h_f \in [0,1]$ | 0.50–0.71 |
| B2 | toric code $-h_f \sum (Z+X)/\sqrt{2}$ | toric $2\times3$, $3\times3$ | gs($h_f = 0.1$) | $h_f \in [0,1]$ | 0.71 |
| B3 | $-\sum_{\square} ZZZZ - h\sum X$ | square $4\times4$ (16) | $\lvert+x\rangle$; gs($h = 1.5$) | $h$ | 0.00–0.95 |
| B4 | $-\sum_{\triangle} ZZZ - h\sum X$ | triangular $6\times3$ (18) | $\lvert+x\rangle$; gs($h = 1.5$) | $h$ | 0.00–0.91 |
| B5 | $-\sum_{\triangle_\uparrow} ZZZ - h\sum X$ | triangular $4\times4$ (16) | $\lvert+x\rangle$; gs($h = 1.5$) | $h$ | 0.00–0.95 |
| B6 | $-\sum ZZ - K\sum_{\square} ZZZZ - 0.5\sum X$ | square $4\times4$ | $\lvert+x\rangle$ | $K \in [0,2]$ | 0.20–0.33 |
| B7 | $-\sum_{\triangle} ZZZ - \lambda\sum ZZ - 0.5\sum X$ | triangular $4\times4$ | $\lvert+x\rangle$ | $\lambda \in [0,2]$ | 0.13–0.33 |
| G4 | $-\sum r^{-6} ZZ - h_f\sum X$ | square $4\times3$ (12) | gs($h_f = 4$) | $h_f$ | 0.00–0.98 |
| R1 | Rydberg, $(\Omega/2)\sum X + \sum V_{ij} n_i n_j$ | square $4\times3$ | $\lvert+x\rangle$ | $\Omega/V_{nn}$ | 0.00–0.45 |
| S1 | toric code $+\epsilon\sum Z$ | toric $2\times3$ | tilted, $\theta = \pi/4$ | $\epsilon$ | 0.63–0.71 |
| I5 | $\sum_{\triangle} ZZZ + h_x\sum X$ | triangular $4\times3$ | $\lvert+x\rangle$ | $h_x$ | 0.00–0.33 |

**Table 1:** Families of the screen (above the line) and two-spin and earlier multi-spin reference families recomputed with the new predictors (below). Sums run over nearest-neighbour bonds ($ZZ$), all triangles ($\triangle$), up-triangles ($\triangle_\uparrow$), square plaquettes ($\square$). gs: ground state of the same Hamiltonian at the given parameter. $\eta$: range over the grid. Grids: 12 points per family (B), 8–13 (references); full lists in `rydberg_screen.py`.

# C Full matrices

Figures 3 to 6 show the main predictors for all 16 rows (families and starts), Fig. 7 the appendix predictors for the highlights. The full appendix matrices are in `rydberg/figs/b2b/appx_1.pdf` to `appx_4.pdf`.

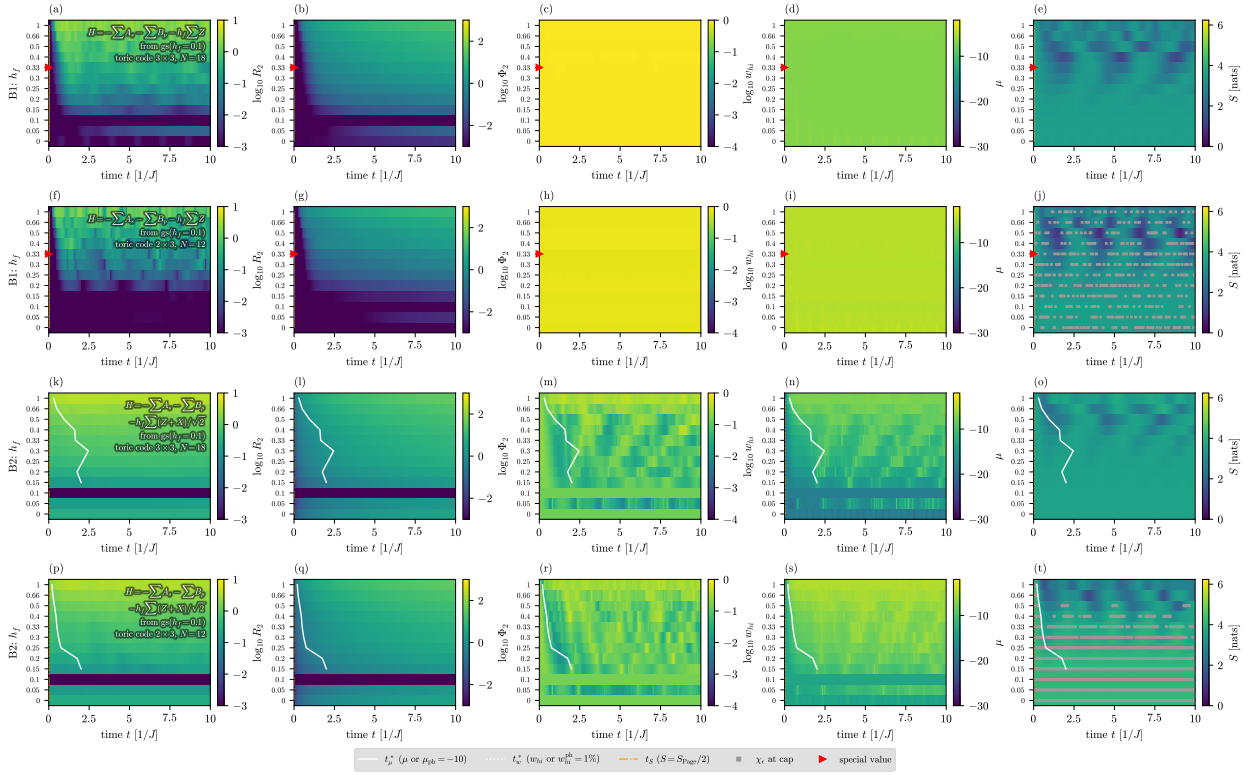


**Figure 3:** Toric code in a $Z$ field (B1) and in a diagonal field (B2), 18 and 12 spins. Layout, lines and colour scales as in Fig. 2. Rows whose run length depends on the control parameter are shown against $t/t_{\max}(g)$. Red triangles: special values (the dual-TFI critical field 0.33 of B1, self-dual points, reductions to earlier families, experimental values of the references). In B1 the state is confined to 256 (32) of $2^{18}$ ($2^{12}$) configurations and all cluster predictors are meaningless (Appendix F).

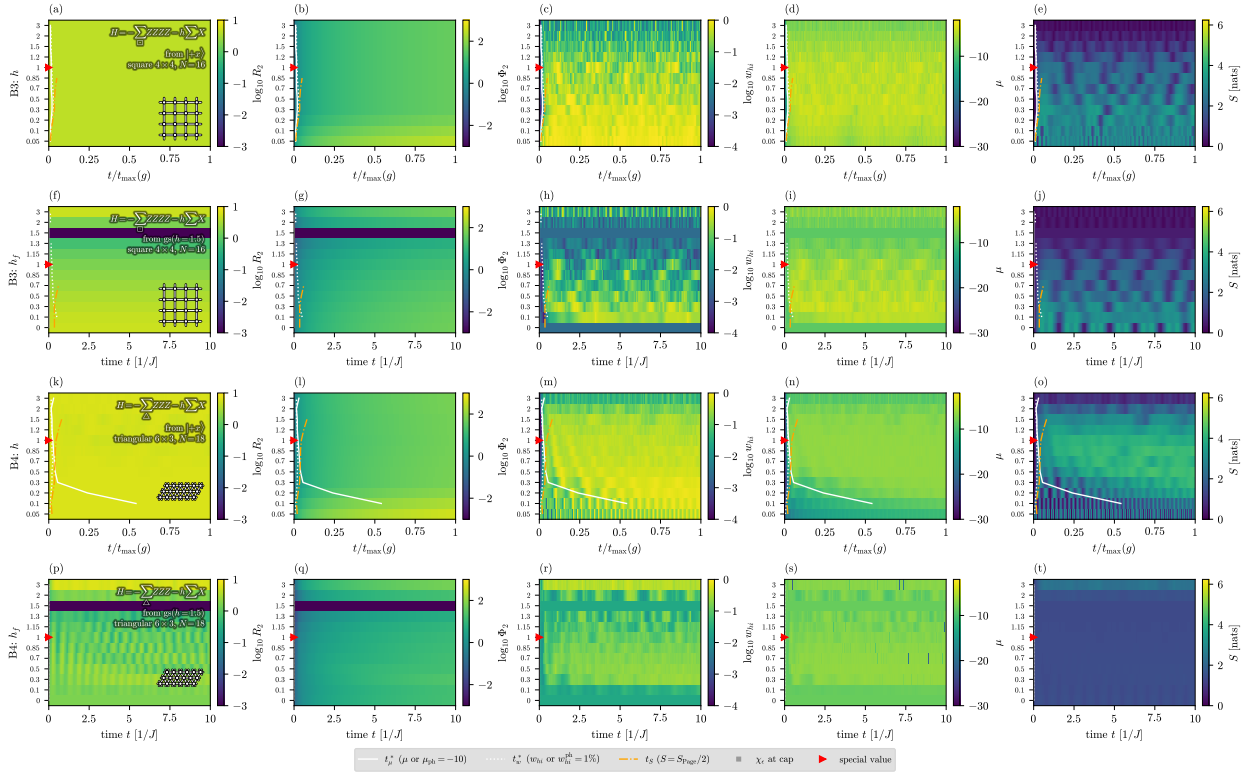



**Figure 4:** Xu–Moore (B3) and Baxter–Wu (B4), each from $|+x\rangle$ and from the ground state at $h = 1.5$. Layout, lines and colour scales as in Fig. 2. Rows whose run length depends on the control parameter are shown against $t/t_{\max}(g)$. Red triangles: special values (the dual-TFI critical field 0.33 of B1, self-dual points, reductions to earlier families, experimental values of the references). The Baxter–Wu ground-state start is an ordered cat state (Appendix F).

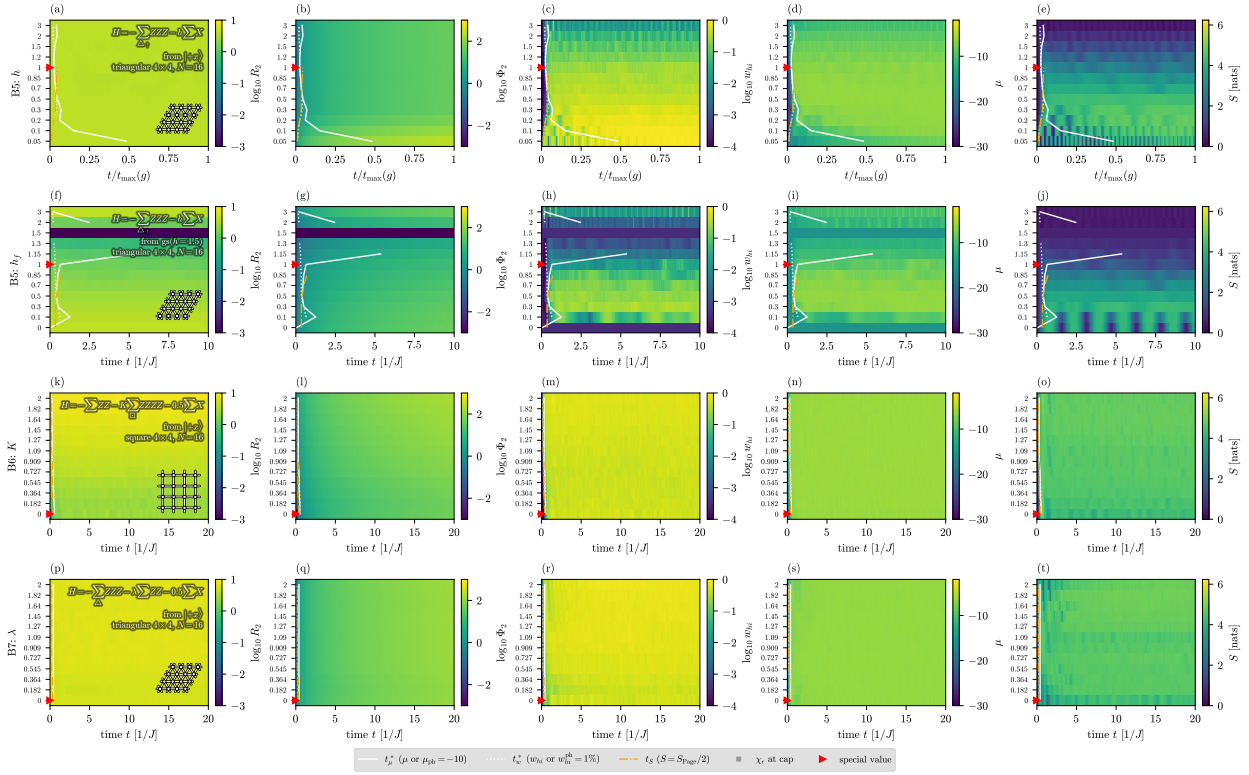


**Figure 5:** Newman–Moore (B5) from $|+x\rangle$ and from the ground state at $h = 1.5$; Ising plus plaquette (B6) and Baxter–Wu plus Ising (B7) from $|+x\rangle$. Layout, lines and colour scales as in Fig. 2. Rows whose run length depends on the control parameter are shown against $t/t_{\max}(g)$. Red triangles: special values (the dual-TFI critical field 0.33 of B1, self-dual points, reductions to earlier families, experimental values of the references).

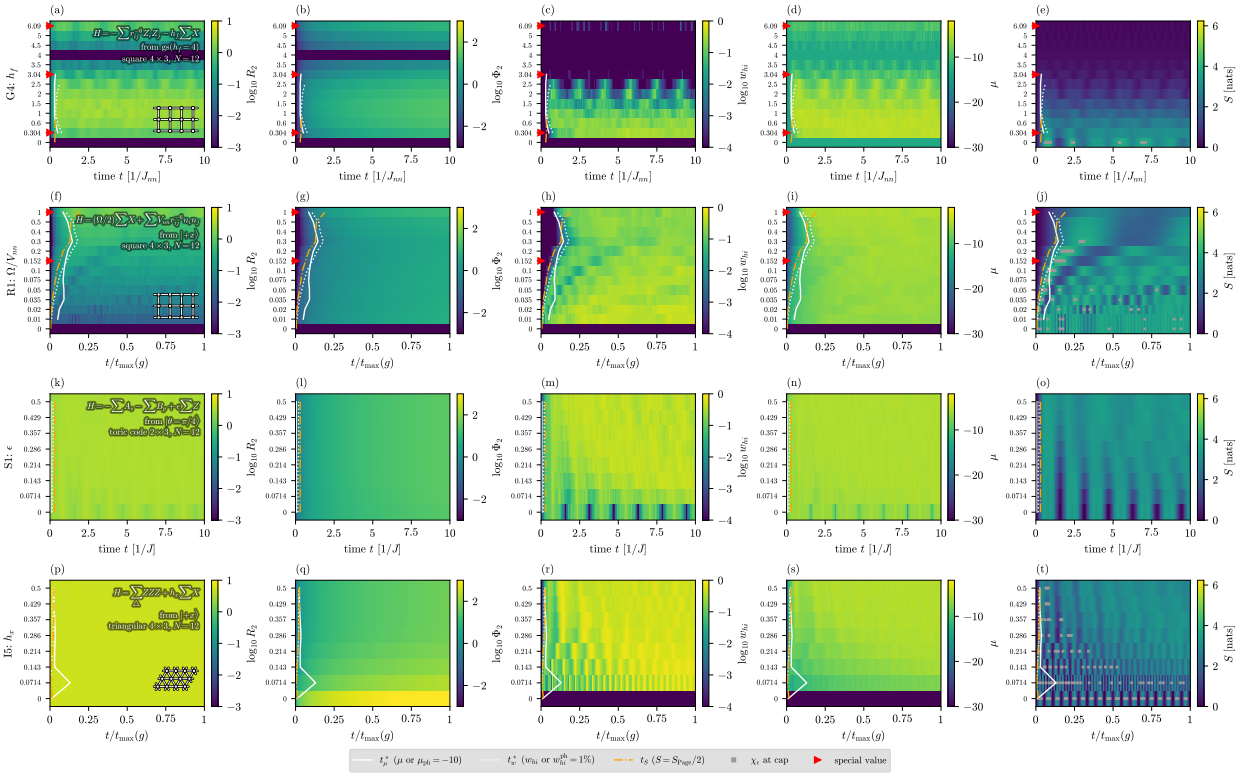


**Figure 6:** Reference families: 2D transverse-field Ising quench from its paramagnetic ground state (G4), Rydberg Ising quench (R1), toric code with a $Z$ field from a tilted product state (S1), three-spin Ising model in a field (I5), all 12 spins. Layout, lines and colour scales as in Fig. 2. Rows whose run length depends on the control parameter are shown against $t/t_{\max}(g)$. Red triangles: special values (the dual-TFI critical field 0.33 of B1, self-dual points, reductions to earlier families, experimental values of the references).

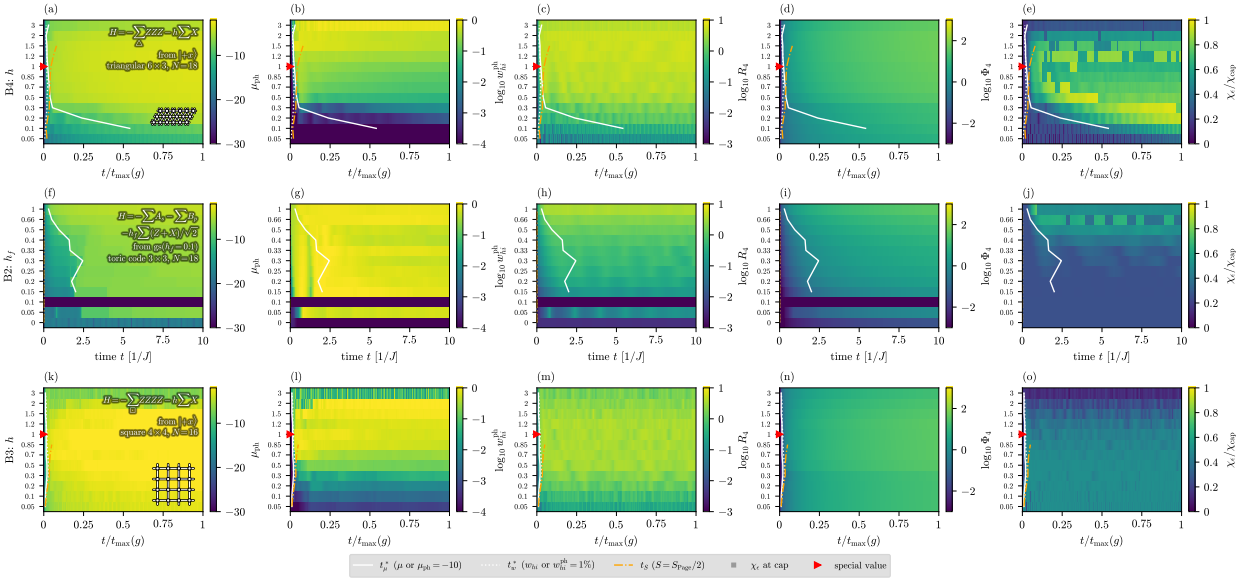


**Figure 7:** Appendix predictors of the three highlighted quenches (rows as in Fig. 2). Columns: phase cumulant level $\mu_{\mathrm{ph}}$, phase high-order weight $\log_{10} w_{hi}^{\mathrm{ph}}$, four-body error rate $\log_{10} R_4$, its integral $\log_{10} \Phi_4$, and the MPS bond dimension $\chi_\epsilon$ (discarded weight $10^{-3}$ at the cut with the largest $S$) as a fraction of its cap, 0 for a product state. Lines as in Fig. 2.

## D Few-body error rate: implementation and checks

**Normal equations in Walsh space.** With $s_A s_B = s_{A\oplus B}$, the Gram matrix of the fit is $G_{AB} = \sum_s p(s) s_{A\oplus B}$, one Walsh–Hadamard transform of $p$, and the right side is the transform of $-i\,\overline{\psi}\,H\psi$ at the clusters $A$. No $2^N \times n_k$ matrix is formed ($n_2 = 172$, $n_4 = 4048$ unknowns at 18 spins). $G$ is real, so the real and imaginary parts are solved with one factorisation (least squares with pivoted QR, relative cut-off $10^{-12}$, since $G$ is singular when $p$ has zeros). The residual is evaluated directly, $|\psi|(v - f_c) = -i(H\psi)e^{-i\arg\psi} - |\psi| f_c$, not as a difference of two large numbers.

**Checks** (asserts in `rydberg2d.py`).

- $R_2 < 10^{-10}$ at all times for a diagonal two-body $H$ (Rydberg interactions without drive).
- $R_2(0) < 10^{-8}$ for the field quench of the 2D transverse-field Ising model from a ground state: there $v(s,0)$ is the constant plus $(1 - h_f/h_0)$ times the two-body classical energy. $R_1(0) > 10^{-3}$ for the same state.
- $R_2$ equals a dense least-squares fit on the explicit basis at 8 spins (random state, random $H$ with a three-spin term).
- Xu–Moore field quench from a ground state: $R_2(0) > 10^{-2}$, $R_4(0) < 10^{-7}$.

**Analytic values seen in the data.** From $|{+x}\rangle$ the velocity at $t = 0$ is $i$ times the classical energy (plus a constant), so $R_2(0)$ is the root of the summed squared couplings of all terms with more than two spins: 6 (Baxter–Wu, 36 triangles), 4 (Newman–Moore, 16 up-triangles; Xu–Moore, 16 plaquettes), $\sqrt{32} = 5.66$ (B7), $4K$ (B6). $R_4(0) = 0$ for every product start and for the ground-state starts whose quench changes only the $X$ field, since all terms have at most four spins; in the toric code in a diagonal field $R_4(0) = 0.002$–$0.03$. In Xu–Moore $R_2(t) = \Delta E$ at all times (Section 6).

## E Check against existing RBM data

*On the two reference quenches with $\alpha = 1$ RBM runs, $\mu$ predicts the infidelity as well as or better than the others; $R_2$ is as good as $\mu$ in the transverse-field Ising quench and much worse in the Rydberg quench* (Table 2).

| quench | $R_2$ | $\Phi_2$ | $\max(w_{hi}, w_{hi}^{\mathrm{ph}})$ | $\max(\mu, \mu_{\mathrm{ph}})$ |
|---|---|---|---|---|
| 2D transverse-field Ising (G4, 1941 points) | 0.91 | 0.88 | 0.91 | 0.91 |
| Rydberg Ising (R1, 2170 points) | 0.51 | 0.83 | 0.47 | 0.93 |

**Table 2:** Spearman correlation between each predictor and $\log_{10} I$ of the $\alpha = 1$ RBM ($I$: infidelity with the exact state), over all grid points and save times $t > 0$; 12 spins, $4 \times 3$ cylinder. RBM data from the previous report (`data/rbm_long.csv`).

**Onset times.** In the transverse-field Ising quench, $t^*_\mu$ is within 0.15 of the RBM failure time $t^*_{\mathrm{inf}}$ (first $I > 10^{-3}$) for $h_f \le 2.5$, and early at the critical and large fields ($t^*_\mu = 0.35$–$0.4$ against $t^*_{\mathrm{inf}} = 2.95$–$3.75$). In the Rydberg quench $t^*_\mu$ is late by a factor 1.1–2.6 for every $\Omega/V_{nn} > 0$ (e.g. 2.7 against 1.65 at 0.1), while $t^*_w$ is early for $\Omega/V_{nn} \le 0.05$ (2.1–3.0 against 2.7–13.5) and late above. Two further reference families of the previous report (dipolar XXZ and long-range Ising) were not recomputed with $R_2$, and the three-spin family I5 changed Hamiltonian (Appendix H), so the check covers two families.

## F Problem cases

**Toric code in a $Z$ field (B1).** The $Z$ field commutes with every plaquette $B_p$ and with the non-contractible $Z$ loops, so the state stays in a sector of $2^N/2^{L_xL_y+1}$ configurations: 32 at 12 spins, 256 at 18. The amplitude is exactly zero elsewhere, so $\log|\psi|$ is dominated by the support indicator and $\mu$, $w_{hi}$ are above threshold at $t = 0$. The fit for $R_4$ has rank equal to the support size and is exact for any $v$ ($R_4 \approx 10^{-11}$), and $R_2$ has rank 127 of the 256 configurations. The model is dual to the 2D transverse-field Ising model [1]; a fair test would use the dual variables, which this screen does not do.

**Ground-state start of Baxter–Wu (B4).** The ground state at $h = 1.5$ on the $6 \times 3$ torus has a gap of only $3.8 \times 10^{-4}$ and $S(0) = 1.40 \approx \ln 4$: this looks like a cat state of the four ordered Baxter–Wu ground states, not the paramagnet intended by the plan. Possible causes, not tested: the transition lies above $h = 1.5$ at this size; the model is not self-dual at $h = 1$ as assumed; or the start state was prepared incorrectly. The gap stayed above the warning level $10^{-4}$, so no warning was printed. This row is not interpreted.

## G Numerical details

- 186 runs (144 new, 42 reference), 44 parallel processes, 2.5 h wall time, 82 CPU hours. Per run: under 1 min at 12 spins, 10–20 min at 16, 53–138 min at 18. The four-body fit dominates at 16 and 18 spins (about 7 s per save time for full rank, 20 s for the rank-deficient B1 fit).
- Peak memory: about 2.8 GB per 18-spin run.
- Ground-state gaps of the pre-quench Hamiltonian: toric code $1.7 \times 10^{-3}$ (12 spins), $2.4 \times 10^{-3}$ (18, $Z$ field) and $1.5 \times 10^{-2}$, $1.2 \times 10^{-3}$ (diagonal field); Xu–Moore 2.4; Newman–Moore 2.3; Baxter–Wu $3.8 \times 10^{-4}$.
- Data in `rydberg/data/b2b/`: raw files per run, gathered files per family, the onset table `summary.csv` and the check tables `validate_rho.csv`, `validate_onsets.csv`.
- Scripts in `rydberg/`: `rydberg_screen.py` (with the environment variable `SCREEN=data/b2b` and the command `run 4`), `b2b_summary.py`, `b2b_validate.py`, and `rydberg_figs.py` with the command `b2b`.

## H Deviations from the plan

The plan is `check_time_evolution/plans/plan_beyond2body.md`.

- *Triangles fixed.* On a triangular torus with three rows, the old `triangles()` also returned the three-site loops that wind around the torus (28 instead of 24 on the $4 \times 3$ torus). It now returns the faces. The reference family I5 therefore has 4 fewer three-spin terms than in the previous report.
- *Toric-code site order* changed to column by column, so the entanglement cut is spatial. The reference S1 has different $S$ than in the previous report.
- $w_{hi}$ is set to 0 for a flat amplitude (Section 2); onset times use the earlier of the amplitude and phase crossing.
- The support size and the rank of the fits were added to the output to detect cases like B1.
- No $R_2$ threshold line: the paper draft with the threshold was not available to check its size rescaling.

- The full matrices are split into four pages of four rows.
- The window is the fraction of save times that are MPS-hard and before the onset, not the time between $t_S$ and the onset as in the plan, because $S$ oscillates around the threshold at weak fields.
- Highlights were chosen by the author of this report, not by the user.

## References


[1] Franz J. Wegner. Duality in Generalized Ising Models and Phase Transitions without Local Order Parameters. *Journal of Mathematical Physics*, 12:2259, 1971.

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[3] R. J. Baxter and F. Y. Wu. Exact Solution of an Ising Model with Three-Spin Interactions on a Triangular Lattice. *Physical Review Letters*, 31:1294, 1973.

[4] M. E. J. Newman and Cristopher Moore. Glassy dynamics and aging in an exactly solvable spin model. *Physical Review E*, 60:5068, 1999.

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